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General Design of the Forming Collar of the Vertical Form, Fill and Seal Packaging Machine Using the Finite Element Method

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The basic part of the vertical form, fill and seal packaging machine is the forming collar. The forming collar provides the shape over which packaging film is smoothly formed at high speed into a cylindrical shape. Describing the forming collar geometry and hence its design is, however, remarkably difficult. This paper presents, for the first time, a flexible methodology for calculating the complete geometry of the film rather than the usually non-complete collar over which the film is formed. That is, a methodology to calculate the film geometry over the collar including the seam along which the film is longitudinally sealed. The film geometry is calculated such that it has minimum deformation energy. Advantages of the proposed methodology include its great flexibility to generate collars with different configurations for different needs. Among the collar generation methods reviewed, the proposed methodology is the first that can systematically consider all collar configuration parameters such as the seam configuration, general package cross-section, flat or straight part of the collar, collar back angle, etc. A means for obtaining the exact collar geometry is also demonstrated. This enables right-first-time and repeatable collar production and reduces the time and cost for producing next generation packaging machines. Copyright © 2010 John Wiley & Sons, Ltd.

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KEY WORDS: forming collar; lap seam; fin seam; vertical form fill and seal machine; developable surface; finite elements method

INTRODUCTION

Most purchased products are delivered in some decorative or protective package. Hence, packaging is essential to most industries.

Among the packages faced in everyday life are bags and pouches used for holding snacks, potato crisps and pasta. These products are packaged in a process schematically illustrated in Figure 1a. As shown in the figure, a flat film (1) of packaging material wound around a reel (2) is drawn at high speed and delivered to the forming collar (3) which smoothly forms it into a cylinder (4) to be filled with a product. From now on, the packaging material film is simply referred to as the *film*. The film-forming process should also bring the outer edges of the film together and overlap them to create a seam (5) along which the film will be sealed. The target product (6) is then dropped into the sealed film which is then cross-sealed (7) and cut to form the complete pouch (8). The cross seal is also the bottom seal for the next pouch. Most commonly the cylinder (4) is of circular cross-section. Figure 1b and c show cross-sections of the film formed inside the cylinder for typical seam shapes. The machine used to perform the process of Figure 1a is called a vertical form, fill and seal machine.

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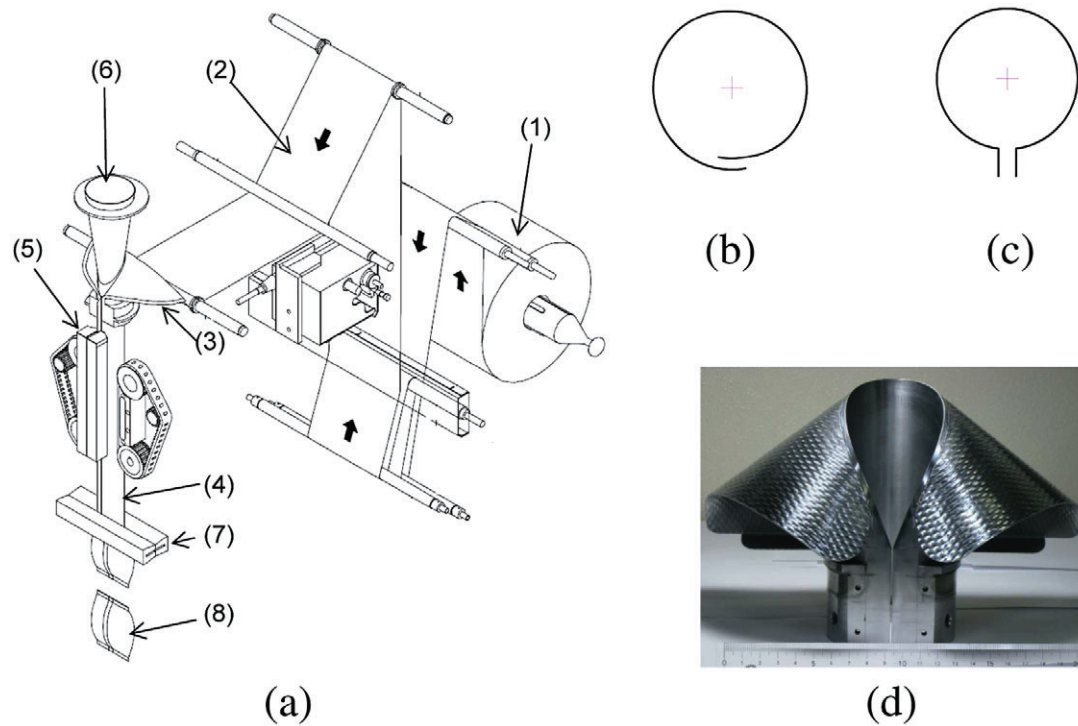


Figure 1. (a) Schematic diagram of a form, fill and seal machine; (b) film cross-section inside the cylinder in the case of a lap seam; (c) film cross-section inside the cylinder in the case of a fin seam; (d) real forming collar.

As shown in Figure 1a the basic part of the vertical form, fill and seal machine is the forming collar (3). The forming collar provides the shape over which the film can be smoothly formed at high speed into a cylindrical shape. However, describing the forming collar geometry and hence its design is remarkably difficult.

The basic concept behind the shape of the forming collar surface is that it can be flattened without distortion (i.e. stretching or compression) onto a plane. That is to say the collar surface must be 'developable'. Figure 2a shows a planar rectangular sheet from which the forming collar will be formed. The width of this sheet is the same as the width of the packaging material. Figure 2b on the other hand shows a typical forming collar, formed without distortion, from this sheet. In Figure 2b, the edge around which the collar is bent is called the *bending curve*. The unrolled version of this curve is shown in Figure 2a and hence is called the *flattened bending curve*. In Figure 2a areas above and below the flattened bending curve are called the *flattened collar* and *flattened cylinder*, respectively. They correspond to the *collar* and *cylinder* surfaces of Figure 2b.

The non-distortion relationship between the surfaces of Figures 2a and b can be mathematically represented by the idea of *isometric correspondence*.¹ Using differential geometry, Culpin proved¹ the existence of two surface classes that satisfy one-to-one isometric correspondence between both the (*flattened collar*, *collar*) and (*flattened bending curve*, *bending curve*) pairs. One of these classes contains the cylinder surface of Figure 2b as well as other surfaces that correspond isometrically to the flattened cylinder of Figure 2a. The other class contains a variety of collar surfaces all corresponding isometrically to the flattened collar surface of Figure 2a. In the following section, the existing methods for generating different surfaces of the collar surfaces class are reviewed.

Theoretical collar generation methods

Early attempts to produce the forming collar. Among early attempts to produce the forming collar was that patented by Monsees *et al.*² He presented a method for physically creating the collar from a

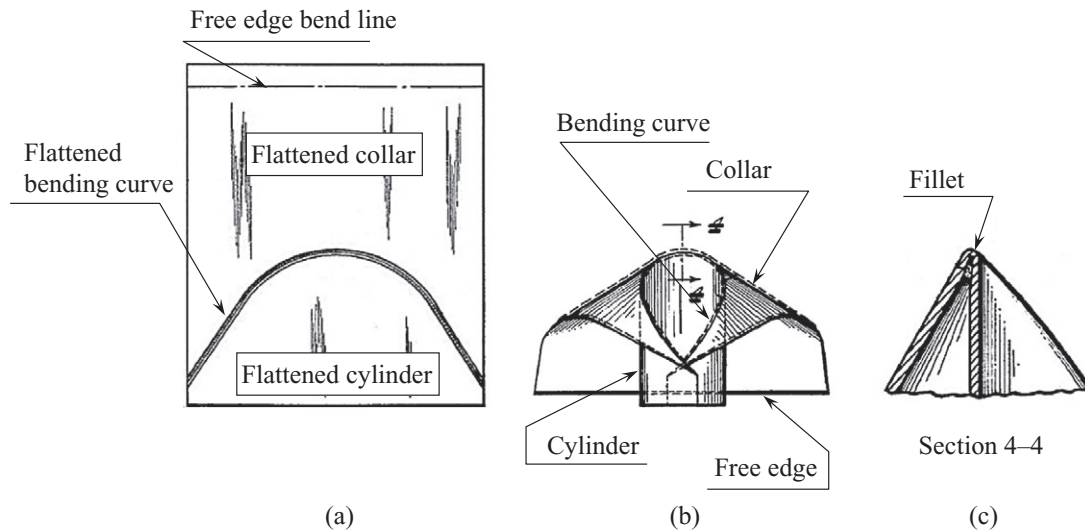


Figure 2. Forming collar of Monsees and Cooper²; (a) flattened forming collar; (b) forming collar; (c) section 4-4.

flat plank like that of Figure 2a. He simply suggested creating a V-groove along the flattened bending curve of Figure 2a. The target of this V-groove is to weaken the bending stiffness and hence facilitate folding around the flattened bending curve. Then by forming the flattened cylinder part of Figure 2a so that it takes the cylindrical shape of Figure 2b, the collar surface will be naturally formed. He also suggested forming a bend along the free edge of Figure 2a to create a perpendicular web. This web stiffens the free edge so that it remains straight. His collar shown in Figure 2b is intended to produce a lap seam (cf. Figure 1b).

However, this method has a number of serious drawbacks that hindered its use for manufacturing real forming collars. For a collar created using this method, the fillet along the bending curve is non-zero as shown in Figure 2c. The problem is that this fillet is not developable since it is not 'ruled surface'. Thus, the collar itself will be distorted (stretched) along the bending curve. Consequently, the film will not slide smoothly over the fillet and hence will have some defects. Therefore, this method is suitable only for very thin planks so that the bend fillet is small enough compared to the film thickness. This will, of course, produce paper-like collars not suitable for a real packaging machine. Another serious drawback of this method is that it does not provide a geometric description of the collar. This also hindered the use of modern manufacturing technology for producing collars.

Forming collar design as a collection of primitive surfaces. Among the first attempts at describing the collar surface geometry was that of Mot,³ who defined the shape of the collar surface using simple geometrical considerations. He represented the collar by a flat triangle centred between two truncated cones. Using this simple approach, he could describe the collar surface geometry for both circular and rectangular (superelliptic) cylinder cross-sections. Later, Zhou *et al.*^{4,5} similarly determined the collar surface geometry for several rounded-corner cylinder cross-sections by using a number of cone and plane surfaces to represent a collar surface. The effect of the pouch seam on the collar geometry was not, however, considered in this approach.

Forming collar design based on classical differential geometry. Based on classical differential geometry Culpin showed¹ that within the collar surface class there is only one surface which is analytic everywhere, that is to say, whose position vector has derivatives of all orders with respect to the parameters specifying a position on the surface. However, he did not describe this surface geometry. Later, Boersma and Molenaar⁶ presented a mathematical description for the collar surface. They also showed that the bending curve must satisfy some condition so that the resulting collar contains a planar triangle. Their model is however limited to a circular cylinder. Later McPherson *et al.*⁷ demonstrated

a method for extending Boersma's model to handle cylinder cross-sections formed from circular arcs and straight line segments. The effect of the pouch seam on the collar geometry is also not considered in this approach.

Collar manufacturing methods

Conventional forming collar manufacturing. The majority of real forming collars are in fact manufactured in a way rather similar to that reviewed in the section *Early Attempts to Produce the Forming Collar*. That is, instead of making the whole collar from a single plate grooved along the flattened bending curve, the plate is instead cut along this curve as shown in Figure 3a. The cylinder part of the collar can then be created by winding the flattened cylinder as shown in Figure 4. However, due to strength and fixing considerations, the cylinder part is made much thicker by machining rather than by forming from a thin flat plate. The flattened-bending-curve edge of the flattened collar is then enforced to follow the bending curve edge of the cylinder as shown in of Figure 4. A real forming collar manufactured in this way is shown in Figure 1d.

As shown in Figure 1d, real collars often have some technical deviations from the ideal manufacturing method described earlier. For example, the collar-cylinder welding is offset from the bending curve to keep it smooth. As another example, the seam region is truncated from the collar. In addition, the back angle θ (shown in Figure 4) is not left free but is forced to take some predetermined value in the real collar. These deviations in turn affect real collar geometry and performance. In fact modern collar design is the result of incremental improvements driven largely by empirical experimentation over the years.^{8,9} For many applications, several collar configurations are produced and tested. The successful collars are chosen and then further refined *by hand*. The hand refinement process involves modifying the collar geometry to improve its performance and in particular, to achieve acceptable tracking at high speeds without damaging the film.¹⁰ This manual design and manufacture process may therefore be considered one of the key barriers against fast and repeatable collar production.

Forming collar computer-aided manufacturing (CAM). In an attempt to overcome these barriers, Hicks *et al.* applied computer-aided design (CAD) and computer-aided manufacturing (CAM) techniques for producing the forming collar.¹⁰ They used the model reviewed in the section *Forming Collar Design Based on Classical Differential Geometry* for the collar geometry and manufactured it using computerized numerical control (CNC) machining. Since the geometric model they used does not produce the pouch seam properly, they modified it *by hand* to create the overlap needed for the seam. Their right-first-time collar showed advantages when it was compared to conventional collars in terms of set-up time, usable film materials, film tension and film slide acceleration.¹⁰

Contribution and organization of this paper

Several sectors have been significantly developed through the use of mathematical modelling and simulations. In fact, computer-aided modelling offered a great tool to improve the understanding and reduce the time of many complex processes. However, the packaging sector generally still did not benefit from this. Actually the underlying design principles of many packaging machines still the result of incremental empirical improvements over the years⁸.

In the section *Forming Collar Computer-aided Manufacturing (CAM)*, it was shown how CAD and CAM can yield right-first-time collars that are promising as compared to conventional collars. The method proposed there is however based on the geometric model reviewed in the section *Forming Collar Design Based on Classical Differential Geometry*, and hence is still limited. That is, it is limited to cylinders with circular cross-sections, the bending curve must satisfy some condition so that the resulting collar contains a planar triangle and it does not generate the overlap necessary for creating the pouch longitudinal seam. Hence, the goal of this paper is to provide an unlimited fully automatic method for collar design to assist the development of the next generation packaging machines.

This paper presents, for the first time, a methodology for calculating a complete and optimized geometric model of the film as well as the forming collar. For a specific bending curve, the optimum collar surface is calculated so that the film exhibits the least deformation when formed over it.

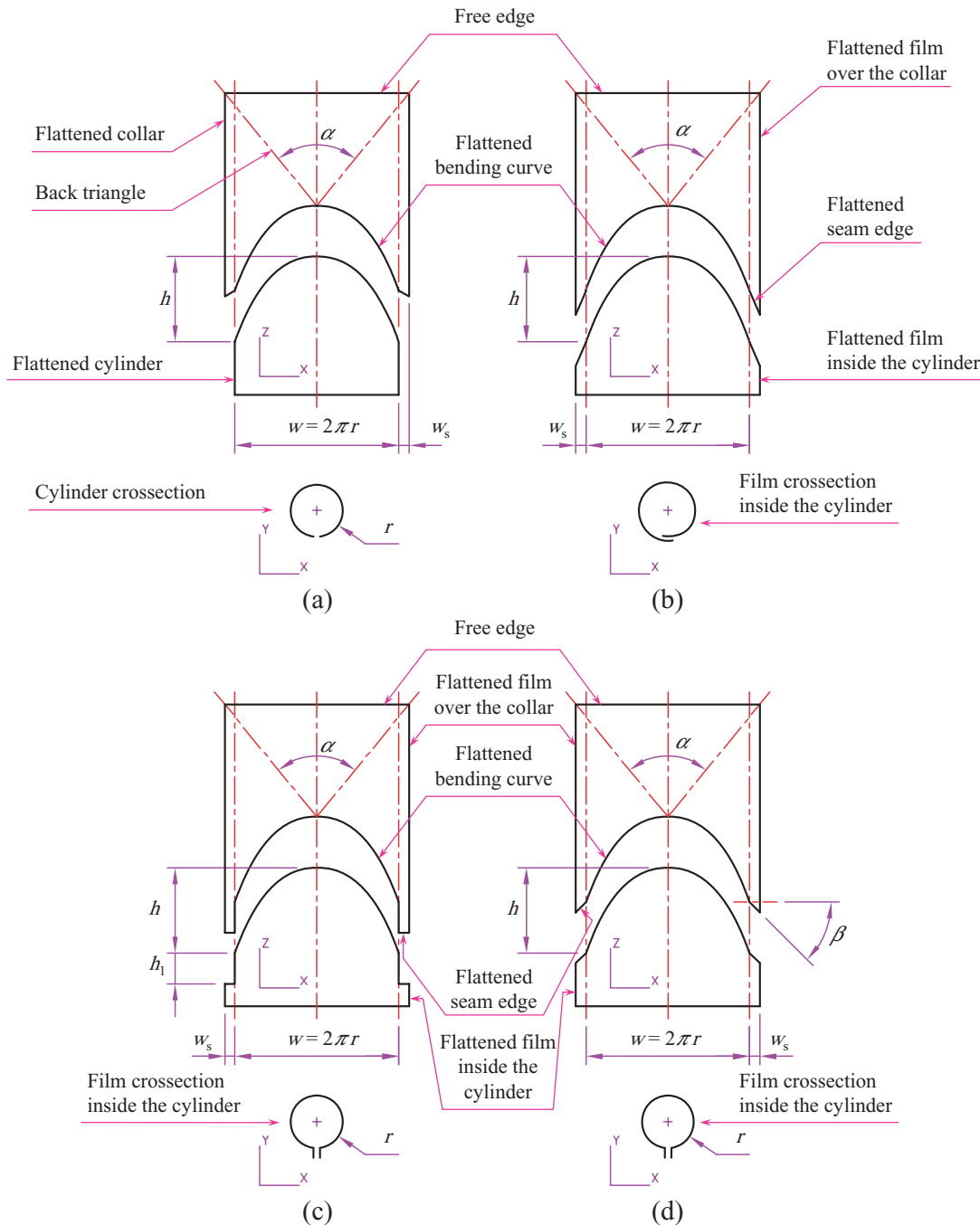


Figure 3. (a) Flattened *non-complete* forming collar along with the cylinder cross-section; (b) flattened film along with the film cross-section inside the cylinder for a pouch with a lap seam; (c) flattened film along with the film cross-section inside the cylinder for a pouch with a fin seam; (d) alternative fin-seam model for which the seam symmetry is assumed to occur along a line with an angle β .

Advantages of the proposed methodology include its great flexibility to generate collars with different configurations for different needs. The methodology can also be fully automated. This enables sophisticated computer-aided engineering (CAE) studies on generated collars for optimized machine material interaction or any target property or performance. Compared with all the methods reviewed,

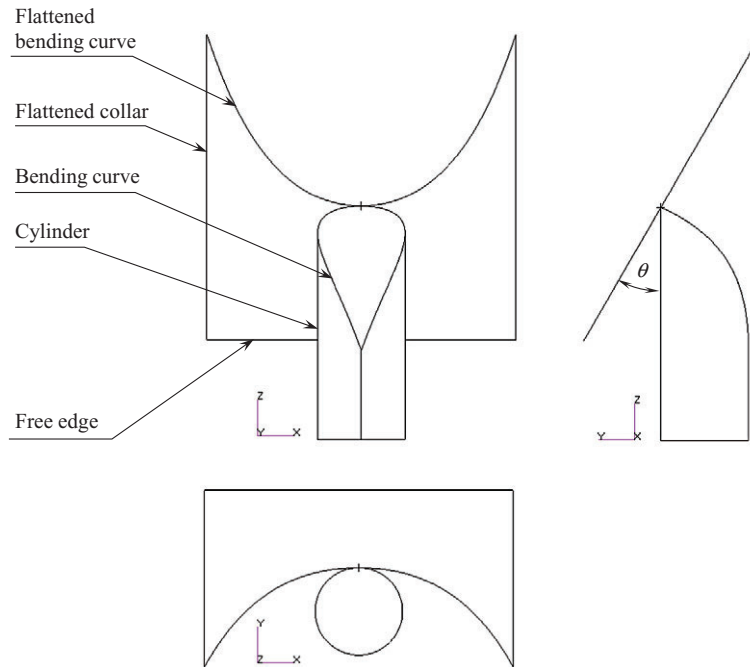


Figure 4. Flattened film to be formed over the forming collar.

the proposed collar design method is the only one that can *systematically* consider all the collar configuration parameters such as seam type, general package cross-section, flat or straight part of the collar, collar back angle θ (cf. Figure 4), etc.

This paper is organized as follows. In the next section, the proposed methodology for collar surface generation is presented. In the section *Application of the Proposed Collar Generation Method*, the proposed method is used to generate five different forming collars. In the section *Verification of the Proposed Collar Generation Method*, the generated collar surface is decomposed into a set of straight lines (ruling lines). Such ruling lines verify the developability of the generated collar surfaces and present a means for exact and repeatable collar manufacturing. Finally, the last section concludes this paper.

FLEXIBLE COLLAR DESIGN USING THE FINITE ELEMENT (FE) METHOD

Regardless of the impediments of the collar generation method reviewed in the section *Early Attempts* . . . , the method is simple, direct and very interesting compared to the other methods. That is, the generated collar surface is not a collection of different primitive surfaces connected to each other, as that of the section *Forming Collar Design as a Collection* . . . , nor a surface satisfying some ad hoc mathematical condition, as that of section *Forming Collar Design Based on Classical Geometry*, it is rather a *naturally generated* smooth surface. In addition, the method does not require a constraint on the bending curve to yield a flat part in the collar as compared to that of the section *Forming Collar Design Based* Alternatively, a flattened part is simply achieved by compelling that part to be so, as was shown in the section *Early Attempts* Moreover, the method takes into account the overlap necessary for producing the seam, which is not even considered by other methods. However, as explained in the section *Early Attempts* . . . , the method as it is produces either defective or paper-like weak collars. Furthermore, since the method provides no geometric description of the collar, it cannot be produced using modern manufacturing technology.

The *naturally generated* collar of the section *Early Attempts* . . . is obtained as the *natural* result of the forming operation described in that section. In fact, this surface is characterized as having the

minimum deformation energy. In this section, a methodology for calculating this natural collar shape will be proposed. This enables avoidance of the drawbacks of the method of the section *Early Attempts*. . . . The Finite Elements Method (FEM) will be the tool used for calculating the collar shape since it intrinsically realizes minimum deformation energy.¹¹ Details are as follows.

Figure 3 shows pieces of the packaging film cut into two pieces along the bending curve. When formed over the forming collar, the upper piece should take the collar shape while the lower piece will take the cylinder shape. For our considered problem, the cylinder surface is generated by winding the flattened cylinder piece around the z -axis to form a cylinder whose cross-section is shown in Figure 3. Different views of this cylinder are shown in Figure 4.

Initial condition

The flattened collar shown in Figure 3 is rotated around the x -axis as shown in Figure 4 so that it makes an angle θ with the cylinder axis. For our simulations, the angle θ is initially set to $\theta_{\text{initial}} = 30^\circ$. This prevents the collar from forming into a surface of the cylinder class explained in the introduction of this paper.

Loading action

Forming the collar is achieved by forcing the *flattened bending curve* of Figure 4 to take the *bending curve* geometry. This pure displacement constraint is the only boundary condition used to form the collar. Rotational degrees of freedom along the bending curve are, however, not restrained. Thus, the collar can freely bend around the bending curve and hence the angle θ of Figure 4 can freely change during the collar forming process.

Film properties

For a well-designed packaging machine, stresses generated in the film during its formation over the forming collar should be low to avoid plastic deformation of the film. Hence, the film material will be assumed linearly elastic with a Poisson's ratio of $\nu = 0.35$. For this material, the generalized Hooke's law yields a stiffness matrix (stress-strain ratio) that is linearly proportional to Young's modulus E ¹². Hence and since enforced displacement is the only loading action used to form the collar (cf. section *Loading Action*), the generated collar geometry will be independent of E .

However, the film thickness, t , does affect the collar geometry. This is because the axial, shear and bending stiffnesses have different proportions to t .¹² This effect will be emphasized later in the section *Application of the Proposed Collar Generation Method*.

Finally it should be noted that though the linearity of the thin shell theory, the collar forming process is non-linear. This is due to the considerable change of the flattened collar of Figure 3 until it takes the final collar shape. This considerable change of the model geometry is against the small deformation assumption of the linear thin shell theory¹³ and hence gives rise to a phenomenon called geometric non-linearity¹⁴. Due to this phenomenon, the load-deformation will not be linearly proportional to each other during the collar forming process. The loading-unloading cycles are, however, elastic, due to the elastic material of the collar.

FE model

As was explained in Figure 2, the forming collar is composed of a cylinder and a collar. The film geometry to be formed inside the cylinder part is completely defined as shown in Figure 4. Hence, it need not be included in the FE model. Thus only the flattened collar of Figure 3 will be modelled using the FE method. In this figure, an extra width w_s is added, which is needed for generating a seam along which the pouch will be longitudinally welded.

The FE model is modelled using MSC Nastran PSHELL thin shell property¹⁵ with membrane, bending and transverse shear stiffness. As shown in Figure 6a this model is mostly composed of

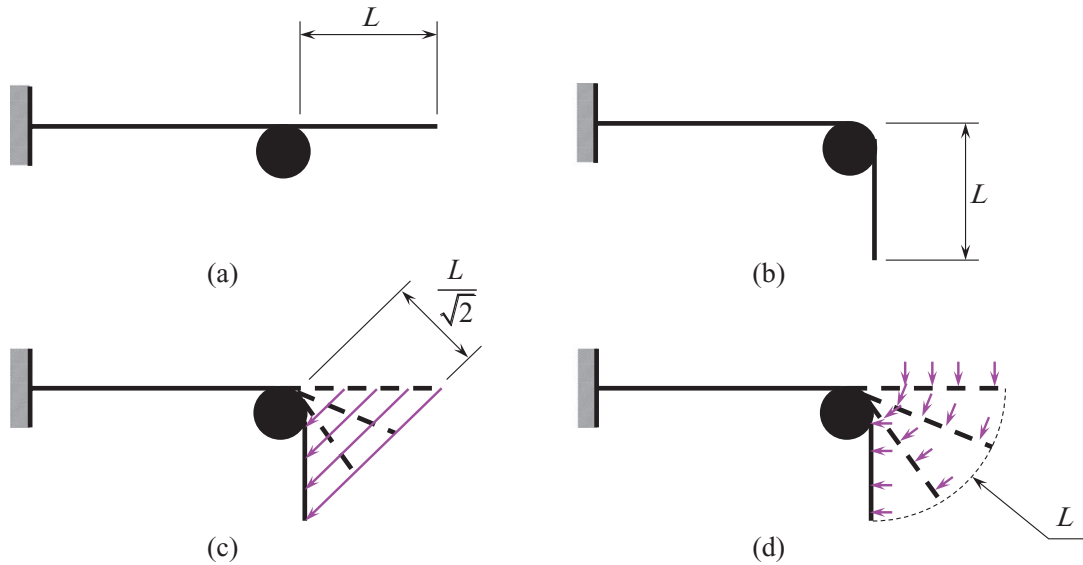


Figure 5. (a) Initial shape; (b) target deformed shape; (c) forming using displacement boundary condition; (d) forming using force boundary condition.

CQUAD4 quadrilateral elements and few CTRIA3 triangular elements of MSC Nastran element library.¹⁵ The mesh density used was varied from fine mesh in areas of high deformation to coarse mesh in areas with less deformation. This variation was carefully performed so that the quality of the elements is maintained. Compared to a finer uniform mesh, this varying-size mesh yielded much reduced calculation time with almost identical nodal results. MSC Nastran's SOL 106 nonlinear static solution sequence is used since it can take account of the geometric non-linearity phenomenon explained in the previous section.¹⁶

Load application strategy

As was explained in the section *Loading Action*, the loading action is an enforced displacement. This displacement is however too large to be directly applied in our FE model. This is illustrated in the following section.

Direct application of enforced displacements; the difficulty. Consider the beam shown in Figure 5a. This beam is to be formed as in Figure 5b. This can be achieved using two options: the displacement boundary condition of Figure 5c or the force boundary condition of Figure 5d.

As shown in Figure 5c, the displacement boundary condition on a point implies enforcing this point to go through a straight line from the original point position to the target end position. For large displacements like the example shown in Figure 5c, it is clear how the enforced displacement boundary condition generates very difficult intermediate shapes and loads, squeezing the beam length from L to $L/\sqrt{2}$.

Referring to our target problem of forming the *flattened collar* shown in Figure 4, Figure 6a shows an isometric view of the FE model of half the *flattened collar* along with the *bending curve*. Also shown in Figure 6a are the straight lines connecting target nodes on the *flattened bending curve* with their corresponding target positions on the final *bending curve*. Similar to Figure 5c, using pure enforced displacement boundary condition constrains every node on the flattened bending curve to move through these straight lines. Under 10 % of the application of the displacement boundary condition, the intermediate shape shown in Figure 6b was obtained. As shown in Figure 6b, the bending curve not only squeezes as was the case in Figure 5c, but the squeeze also induces the considerable collar wrinkling shown in Figure 6b. Similar to Figure 5c, the squeeze and wrinkle gradually increase with

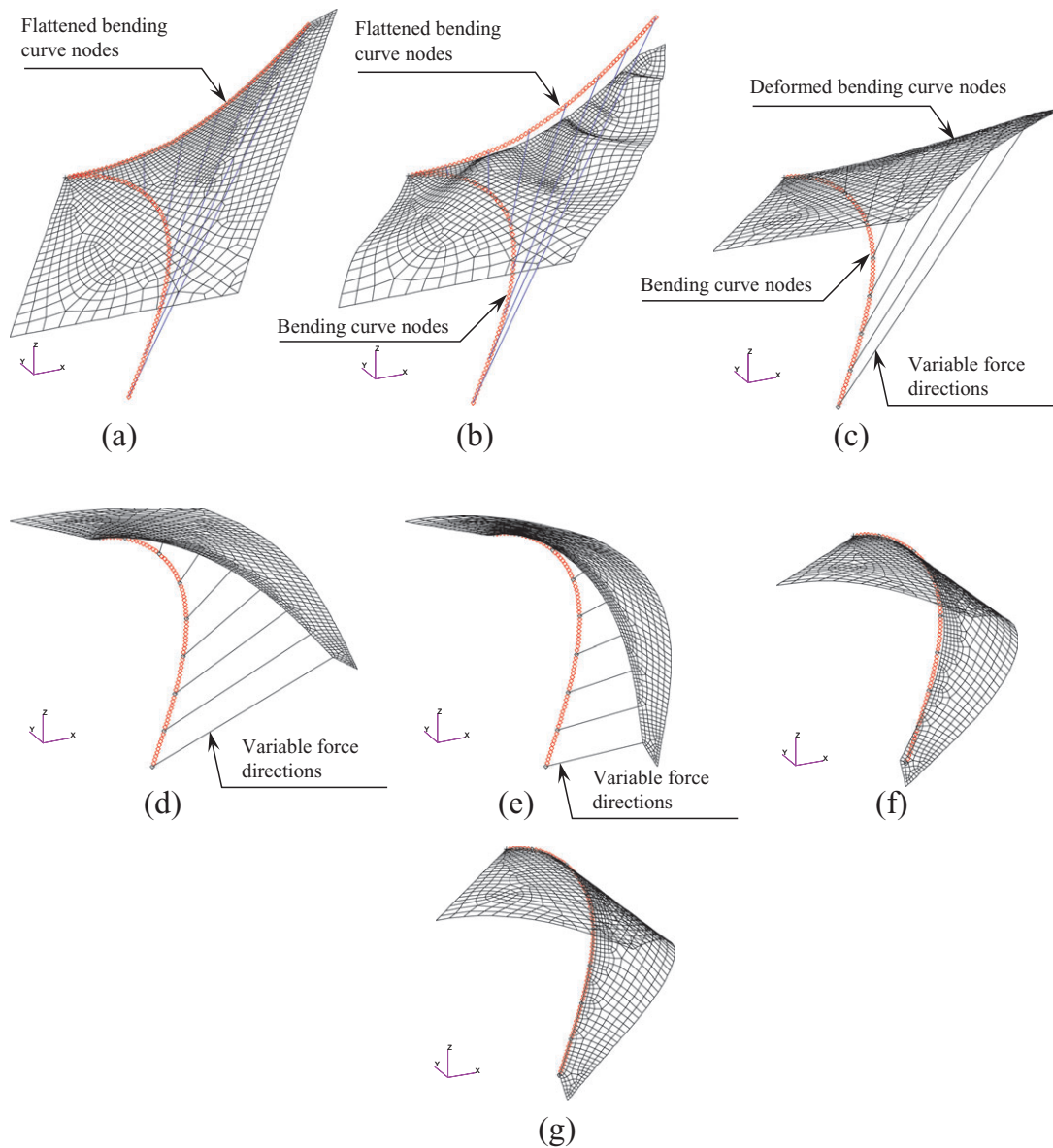


Figure 6. Load application strategy; (a) flattened collar along with the lines connecting the flattened bending curve nodes with their corresponding nodes on the bending curve; (b) collar shape due to the application of 10% of pure displacement boundary condition; (c) collar formed under 3% of the force boundary condition; (d) collar formed under 10% of the force boundary condition; (e) collar formed under 20% of the force boundary condition; (f) collar formed under 100% of the force boundary condition; (g) collar shape formed using the displacement boundary condition subsequent to the force boundary condition.

the application of enforced displacements to a maximum at half the application of such enforced displacements. Then, conversely, they begin to be gradually relieved until vanishing completely at full application of the enforced displacements. In trying to simulate this adverse process, soon after the stage shown in Figure 6b, MSC Nastran failed to calculate subsequent stages and the analysis was unsuccessfully terminated.

Achieving large enforced displacements. Referring again to the simple forming example of Figure 5, as opposed to the adverse scenario of Figure 5c, Figure 5d shows how large deformations can be

smoothly achieved using force boundary condition. The difficulty however is that the magnitudes of the forces required to achieve such deformation are unknown and are very difficult to calculate due to the geometric non-linearity phenomenon explained in the section *Film Properties*. In addition, to achieve large deformation smoothly, the directions of the forces should change according to the deformed position as shown in Figure 5d. This further complicates the calculation of the forces required to achieve the target enforced displacement. In the following, a strategy is proposed to circumvent this difficulty by using a combination of force and displacement boundary conditions to exactly achieve the target enforced displacement.

Stage 1: Force boundary condition. Figure 6a shows the flattened collar along with the bending curve. Since it is required to pull the flattened bending curve to take bending curve geometry, a set of *variable-direction* pulling forces are distributed over all the flattened bending curve nodes. At the start of applying these forces, every force points from a node on the flattened bending curve to its corresponding final location on the bending curve. Some of the directions of these forces are shown in Figure 6a. On gradual application of these forces, the flattened bending curve is gradually pulled towards the bending curve and simultaneously the directions of these pulling forces change accordingly so that they keep pointing from the deformed bending curve to the fixed bending curve as shown in Figure 6c–e. Under full application of the pulling forces, the initially flattened bending curve becomes very close to the bending curve as shown in Figure 6f. Regarding the magnitudes of the pulling forces, they are held constant and uniform all over the bending curve nodes. Suitable value for the force magnitude was chosen by trial.

Stage 2: Enforced displacement boundary condition. In the previous loading stage, the flattened bending curve was pulled close enough to the bending curve as shown in Figure 6f. Thus, a displacement boundary condition can be applied to achieve the remaining small deformation towards the bending curve to obtain the collar shape shown in Figure 6g.

APPLICATION OF THE PROPOSED COLLAR GENERATION METHOD

In this section, the methodology proposed earlier in the paper will be applied to forming collar generation. It is shown how a wide variety of collar configurations can be systematically obtained by applying different boundary conditions. During this section, the following dimensions for the film blank of Figure 3 will be used: $w = 2\pi r = 200$ mm, $w_s = 12.5$ mm and $\theta = 90^\circ$. The bending curve used in these simulations is a 4th order polynomial given by

$$y(x) = -342x^4 - 7.099x^2 \quad (1)$$

and yields h/r ratio of ≈ 3.3 . This curve has a continuous 3rd order derivative in contrast to the constraint derived in Boersma and Molenaar⁶ for obtaining a flat triangle in the collar reviewed in the section *Forming Collar Design Based . . .*

Unconstrained collar

An unconstrained collar means that the collar will be generated under the loading action explained in the section *Loading Action* without any additional constraint. For this collar, a film with the configurations of Figure 3a is used. Due to the symmetry of the collar and the cylinder cross-section, only the right half of the flattened collar of Figure 3a is modelled along with the symmetry boundary conditions (refer to Figure 8, Table 1 and Table 2 for detailed descriptions of the boundary conditions). Using the methodology proposed in the section *Achieving Large Enforced Displacements*, this collar is calculated and displayed in Figure 7a. This collar was generated using a film gauge (thickness) of $t = 0.1$ mm. It was found that increasing the thickness up to 1 mm yields a slight change in the collar geometry.

Table 1. Boundary condition details of the collar FE model.

Boundary condition	Affected region (refer to Fig. 8)	Description (refer to Fig. 8)
Fix_base_node	Base node	U_x , U_y and U_z fixed
Temporary_fix_base_node	Base node	R_x fixed
Fix_tip_nodes	Tip nodes	U_x , U_y , U_z , R_x , R_y and R_z fixed
Symmetry_BC	Symmetry nodes	U_x , R_y and R_z fixed
Bending_curve_follower_forces_A	Bending curve nodes	Follower forces with constant magnitudes and variable directions always pointing from the moving bending curve nodes towards the corresponding tip nodes. For $t = 1$ mm, force magnitude of 10 N was used. For other thicknesses, magnitude is proportional to t^3 .
Bending_curve_enforced_displacements_A	Bending curve nodes	Enforced U_x , U_y , U_z forcing every bending curve node to coincide with its corresponding tip node
Bending_curve_follower_forces_B	Seam edge nodes	Follower forces with constant magnitudes and variable directions always pointing from the moving seam edge nodes towards the corresponding tip nodes For $t = 1$ mm, force magnitude of 300 N was used. For other thicknesses, magnitude is proportional to t^3 .
Bending_curve_enforced_displacements_B	Seam edge nodes	Enforced U_x , U_y , U_z forcing every bending curve node to coincide with its corresponding tip node
Fin_seam_symmetry_BC	Seam edge nodes	$R_y = 0^\circ$ and $R_z = -90^\circ$

Table 2. Boundary conditions applied in every solution step.

Boundary condition (refer to Table 1)	Step 1	Step 2	Step 3 ^a	Step 4 ^a	Step 5 ^a
Fix_base_node	✓	✓	✓	✓	✓
Temporary_fix_base_node	✓				
Fix_tip_nodes	✓	✓	✓	✓	✓
Symmetry_BC	✓	✓ ^b	✓	✓	✓
Bending_curve_follower_forces_A	✓				
Bending_curve_enforced_displacements_A		✓	✓	✓	✓
Bending_curve_follower_forces_B			✓		
Bending_curve_enforced_displacements_B				✓	✓
Fin_seam_symmetry_BC					✓

^a Applied only for collar of the section *Collar with Fin Seam*.

^b Not applied for collar of the section *Collar with Lap Seam and General Cross-section*.

Collar with straight edge

The *free edge* shown in Figure 3a is now enforced to be straight. In the FE model, this is accomplished by attaching an infinitely rigid beam along the free edge of the initially flat collar of Figure 4. The RBE2 element of MSC Nastran's element library was used for this purpose.¹⁵ Using a film thickness of 0.1 mm as in the previous section, the collar obtained under a 60% application of the enforced displacement boundary condition described in stage 2 of the section *Achieving Large Enforced Displacements* . . . is calculated and displayed in Figure 7b. By comparing Figure 7a and b, it is observed that constraining the collar free edge to be straight reduces the back angle θ . In addition, it is observed from Figure 7b that a region of the collar cripples. This reveals that the 0.1 mm film used yields a collar that is too weak to pull the straight edge and reduce the back angle θ . In fact, after the stage shown in Figure 7b, MSC Nastran failed to calculate subsequent stages and the analysis was

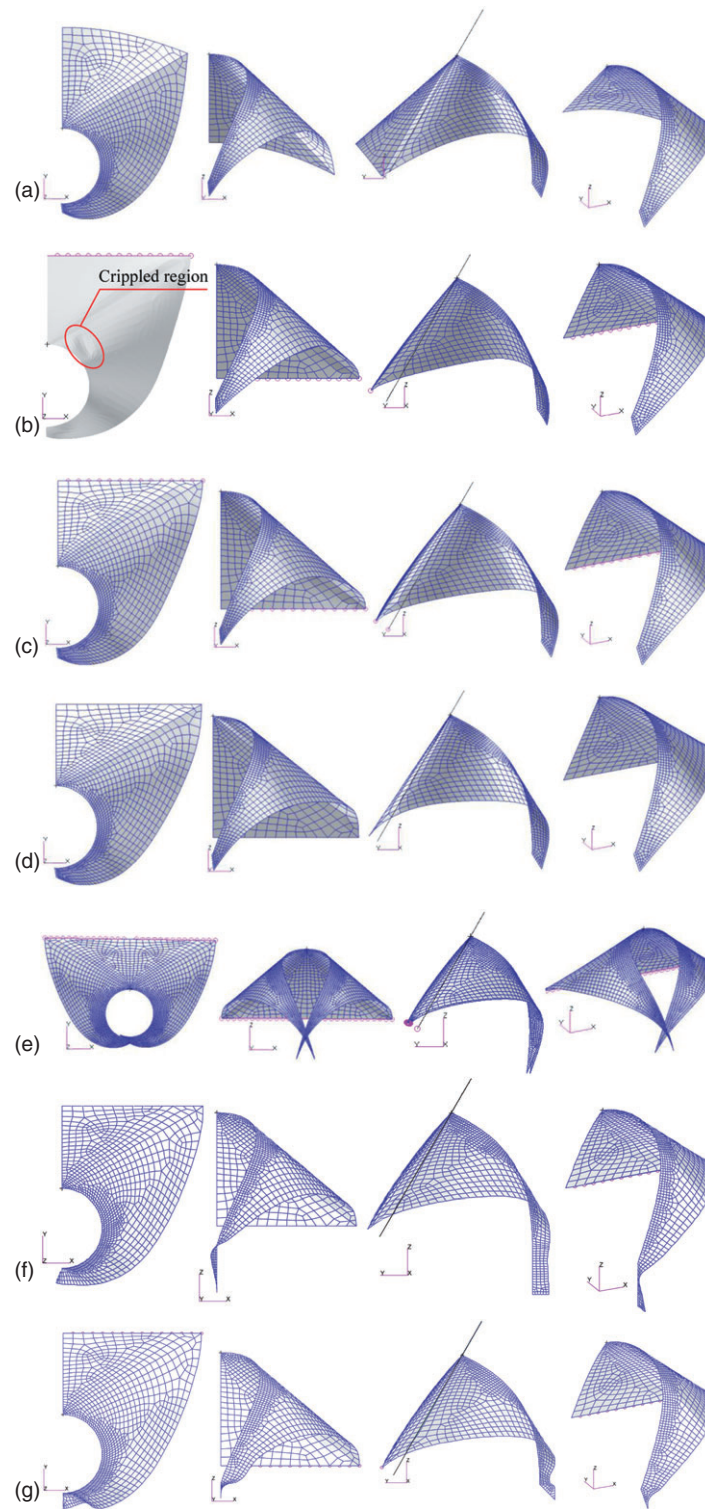


Figure 7. Various forming collars generated using the proposed methodology. (a) Unconstrained collar, $t = 0.1$ mm; (b) collar with straight edge, under 60% application of stage 2 enforced displacement, $t = 0.1$ mm; (c) collar with straight edge, $t = 0.5$ mm; (d) collar with flat triangle, $t = 1$ mm; (e) collar with straight edge and lap seam, $t = 1$ mm; (f) collar with straight edge and fin seam, $t = 1$ mm; (g) collar with straight edge and fin seam, $t = 1$ mm (assuming seam symmetry occurs along a line with angle β , c.f. Figure 3d).

unsuccessfully terminated. To counteract this, the film thickness was increased to 0.5 mm to strengthen the collar and obtain the crippling-free collar displayed in Figure 7c.

Collar with flat triangle

Instead of constraining the *free edge* to be straight as in the previous section, the *back triangle* of Figure 3a is constrained to be flat. Similar to the previous section, this is accomplished by attaching an infinitely rigid RBE2 element to all nodes enclosed by this triangle in the initially flat collar shown in Figure 4. The calculated collar is displayed in Figure 7d. As mentioned earlier, the used bending curve of Equation 1 has a continuous 3rd order derivative in contrast to the condition reviewed in the section *Forming Collar Design Based. . . .*

By comparing Figure 7c and d, it is observed that constraining the collar back triangle flat further reduces the back angle θ . In fact, using 0.5 mm film thickness as in the previous section yielded crippling similar to that shown in Figure 7b. Hence, the film thickness was increased again to 1 mm to further strengthen the collar and obtain the crippling free collar displayed in Figure 7d.

Collar with lap seam and general cross-section

As an example of the flexibility of the proposed methodology, it is used to generate a collar with a non-circular and non-symmetric cross-section as explained in the following.

Consider a collar that generates a lap-type seam as shown in Figure 1b. This is usually achieved by using a circular cross-section collar.¹⁰ A symmetric circular cross-section however does not properly produce the overlap necessary for making the lap seam. Hence, the collar has to be modified manually to produce such overlap. This modification is performed on a trial and error basis.¹⁰ Such manual modification could have been bypassed by using an appropriate collar cross-section such as that shown in Figure 1b. Thanks to the flexibility of the proposed collar generation methodology, it will be applied here for generating a collar with the general non-circular non-symmetric cross-section shown in Figure 1b. This cross-section can be represented by an Archimedean spiral¹⁷ given by the following equation.

$$r = a\theta, : \theta \in [\theta_1, \theta_2] \quad (2)$$

This spiral is characterized by a constant separation distance. For the considered problem the separation distance is chosen as 2 mm, the spiral length is constrained to have the length $w + 2w_s$ and the overlap angle is set as 45° . These yield $a = 0.00031831$, $\theta_1 = 96.461$ radians and $\theta_2 = 103.53$ radians. This curve is further rotated and moved so that its midpoint coincides with the origin as displayed in Figure 1b.

The film of Figure 3b is used for this collar. Since the cylinder cross-section of Figure 3b is not symmetric, the complete FE model of the collar is needed, rather than the half model used for previous symmetric collars. Detailed descriptions of the boundary conditions are given in Figure 8, Table 1 and Table 2. Using a film thickness of 1 mm, this collar is calculated and displayed in Figure 7e.

Collar with fin seam

As another example of the flexibility of the proposed methodology, it is used to generate the film geometry over a collar that produces the fin seam shown in Figure 1c.

To achieve this, the flattened bending curve and the film cross-section of Figure 3c are used. Since the film and the collar cross-section are symmetric, only half the collar is modelled. Following the methodology proposed in the section *Flexible Collar Design Using the Finite Element . . .*, the flattened film inside the cylinder of Figure 3c is wound around the z -axis so that it takes the cross-section shown in the same figure. Note that the flattened seam edge (which is a part of the flattened bending curve) is now perpendicular to the flattened bending curve. Due to this abrupt change, the seam edge is treated separately from the bending curve. That is the two-stage methodology of the section *Achieving Large Enforced Displacements* is firstly applied to enforce the flattened bending

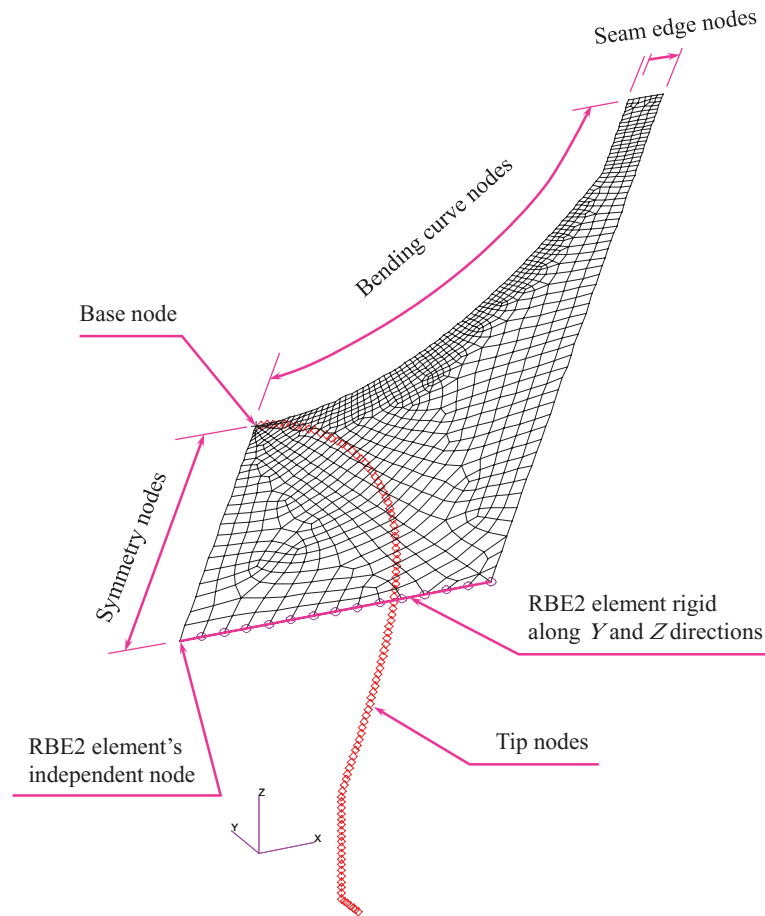


Figure 8. Regions upon which boundary conditions are applied.

curve (excluding the flattened seam edge) of Figure 3c to follow its corresponding bending curve geometry, cf. Figure 4. These two stages are denoted by Step 1 and Step 2 in Table 2 and their resulting collar shapes are shown in Figures 9a and b. Subsequently the methodology is repeated again to enforce the initially flattened seam edge to take its corresponding seam edge shape through the two stages denoted by Step 3 and Step 4 in Table 2 and whose resulting collar shapes are shown in Figures 9c and d. Finally, one more boundary condition is applied along the seam edge. This is the symmetry boundary condition ($R_Y = 0^\circ$ and $R_Z = -90^\circ$), which is applied in Step 5 of Table 2 to obtain the final collar shape shown in Figure 9e and Figure 7f. All boundary conditions used are reported in Table 1 and Figure 8 for reference.

As seen from Figure 7f, the upper seam region is found to exceed the collar plane of symmetry despite the symmetry boundary condition applied in Step 5 of Table 2. In a real collar with a bending curve similar to that of Figure 3c, the two opposing fin seam plies were however found to push each other and reach equilibrium at configurations as shown in Figure 10. This upper seam region was however found to be an unstable region that fluctuates between different configurations as shown typically in Figures 10a and b.

To obtain the fin seam shape in perfect equilibrium in the FE model, one more boundary condition is needed. This boundary condition is a contact surface along the collar plane of symmetry that prevents the fin seam from exceeding that plane. However, since activating contact analysis in FE modelling considerably increases the computational time and resources, another simpler and approximate method was followed. That is the symmetric contact of the two seam plies was assumed to occur along the line with the angle θ shown in Figure 3d. This can be achieved simply by truncating the flattened film of Figure 3c along that line as shown in Figure 3d. As shown in Figure 3d, this line now

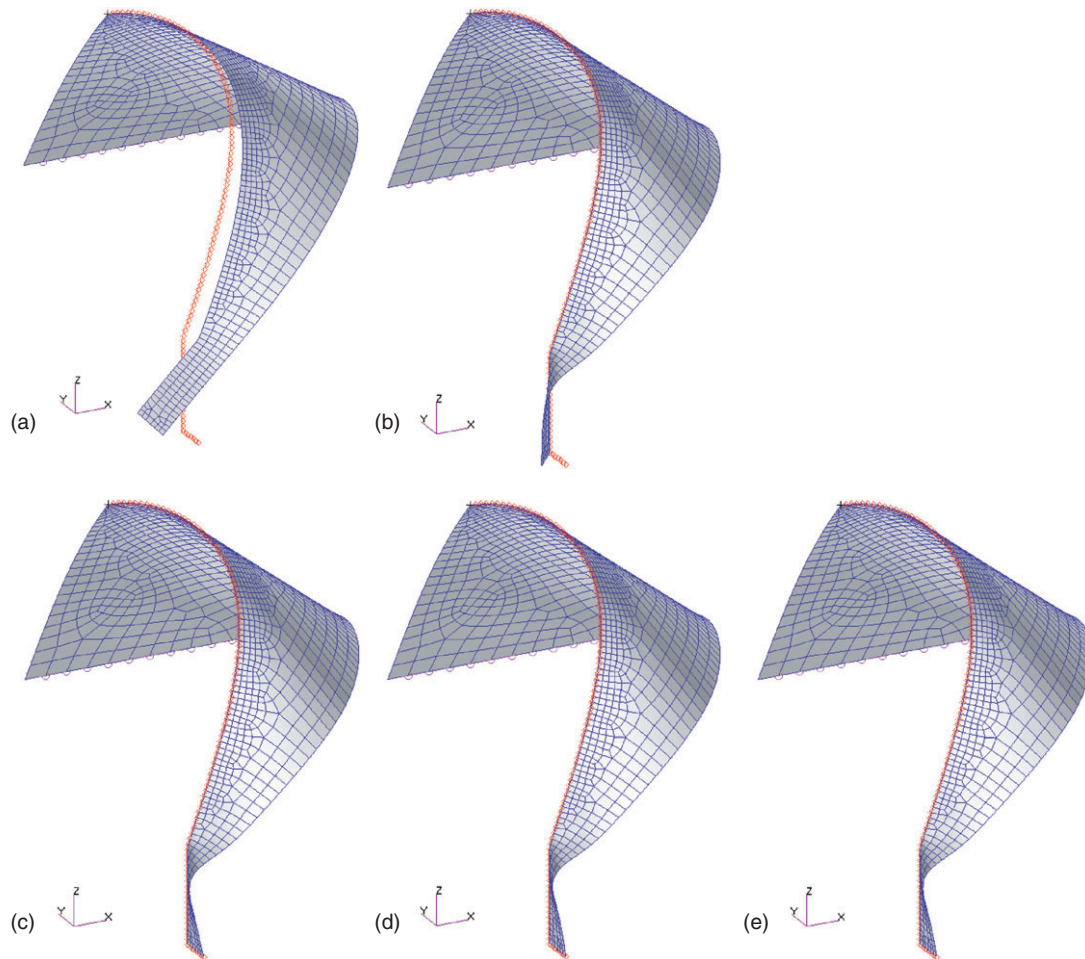


Figure 9. Collar generated in the section *Collar with Fin Seam* at the end of (a) solution step 1; (b) solution step 2; (c) solution step 3; (d) solution step 4; (e) solution step 5.

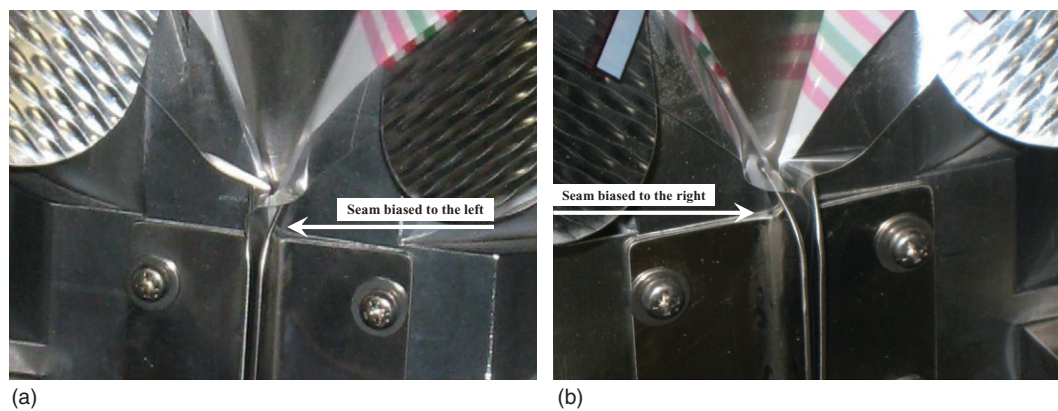


Figure 10. Typical configurations for the unstable upper seam region for a real fin seam collar (courtesy of Kawashima Packaging Machinery LTD. Tokyo, Japan, www.kawashima-pack.co.jp).
(a) Seam biased to the left; (b) seam biased to the right.

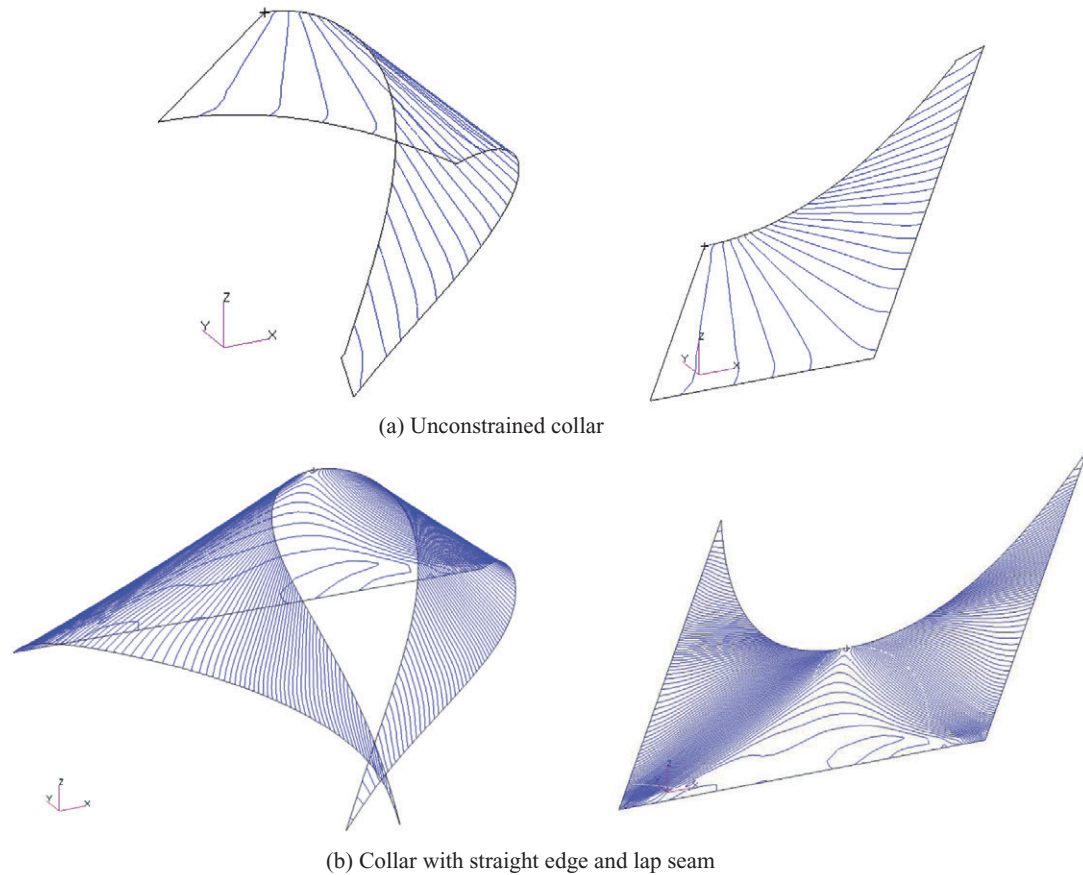


Figure 11. Preliminary ruling lines estimates as the contours of constant rotation vector magnitude.

replaces the perpendicular flattened seam edge of Figure 3c. Thus, the same procedure used to generate the collar of Figure 7f is repeated to obtain the perfectly symmetric collar shown in Figure 7g for which $\theta = 45^\circ$. For both the collars of Figures 7f and g, a film thickness of 1 mm was used.

As was shown in the current and previous sections, the complete film geometry over the collar (including the seam) could be systematically calculated for the first time. This clearly demonstrates the flexibility and strength of the proposed methodology as compared to all other collar generation methods reviewed in the introduction of the paper.

VERIFICATION OF THE PROPOSED COLLAR GENERATION METHOD

Since the collar surface is developable, it is a *ruled surface*.¹⁸ As a consequence, the collar surface can be defined in terms of a set of straight lines called *ruling lines*.¹⁹ As a verification of the proposed collar generation method, the ruling lines of the generated collars are to be calculated.

Such ruling lines can be calculated from meshes representing developable surfaces by computing the geodesic circles as was explained in Kilian *et al.*²⁰ However, ruling lines are obtained here directly from the FE model results. In other words, ruling lines are obtained as the contours of constant rotation vector magnitudes as displayed in Figure 11. In this figure contours are calculated by interpolating rotation magnitudes at every node. Hence, they are subject to interpolation errors. Despite interpolation errors, calculated contours are nearly straight all over the collar except for a narrow region along the collar outer edges and the nearly flat triangular region at the straight edge of the collar of Figure 11b. Considering the collar outer edges, interpolation is known to have high errors due to the edge effect. This phenomenon is known as the Gibbs phenomenon and is explained in Oppenheim and Schafer.²¹

On the other hand, since the triangular region at the straight edge of the collar of Figure 11b is nearly flat, ruling lines in this region are not uniquely defined. This explains why the ruling lines in this region are not straight. After correcting the ruling lines of Figure 11, they can be used to generate a perfect collar surface or can be input directly to a CNC machine to obtain a smooth collar ready for operation.

CONCLUSIONS

Several sectors have been significantly developed through the use of mathematical modelling and simulations. In the packaging sector however, the underlying design principles of many packaging machines are still the result of incremental empirical improvements over the years.

This paper presents, for the first time, a flexible methodology for calculating the complete geometry of the packaging film rather than the usually non-complete collar over which the film is formed. The film geometry is calculated so that it has minimum deformation energy. Advantages of the proposed methodology include its great flexibility to generate collars with different configurations for different needs. Moreover, it enables sophisticated CAE studies for optimized machine material interaction or any target property or performance. Among the collar generation methods reviewed, the proposed methodology is the first that can *systematically* consider all collar configuration parameters such as seam configuration, general package cross-section, flat or straight part of the collar, collar back angle, etc. A means for obtaining the exact collar geometry is also demonstrated. This enables right-first-time and repeatable collar production and reduces the time and cost for producing next generation packaging machines.

REFERENCES

1. Culpin D. A metal-bending problem. *Mathematics Scientist* 1980; **5**: 121–127.
2. Monsees CE, Cooper FH. Tube former and method of making same. United States Patent Office, No. 2940408, 1960.
3. Mot E. The 'shoulderproblem' of forming, filling and closing machines for pouches. *Applied Sciences Research* 1973; **27**: 1–13.
4. Zhou Y-J, Wang Z-W. Study on rhomb shoulders in packaging machines. *Packaging Technology and Science* 2004; **17**: 287–294.
5. Zhou Y-J, Qian X-M. Non-centrosymmetric section forming shoulders in packaging machines. *Packaging Technology and Science* 2006; **19**: 97–104.
6. Boersma J, Molenaar J. Geometry of the shoulder of a packaging machine. *SIAM Review* 1995; **37**(3): 406–422.
7. McPherson CJ, Mullineux G, Berry C, Hicks BJ, Medland AJ. Design of forming shoulders with complex cross-sections. *Packaging Technology and Science* 2005; **18**: 199–206.
8. Hicks BJ, Medland AJ, Mullineux G. A constraint based approach to the modeling and analysis of packaging machines. *Packaging Technology and Science* 2001; **14**(5): 209–226.
9. McPherson CJ, Mullineux G, Berry C, Hicks BJ, Medland AJ. The performance envelope of forming shoulders and implications for design and manufacture. *Proceedings of the Institution of Mechanical Engineers-Part B: Journal of Engineering Manufacture* 2004; **218**: 925–934.
10. Hicks B, Mullineux G, Matthews J, Medland T. Towards an integrated CAD/CAM process for the production of forming shoulders with exact geometry. *Proceedings of the Institution of Mechanical Engineers-Part B: Journal of Engineering Manufacture* 2007; **221**: 1521–1531.
11. Hutton DV. *Fundamentals of Finite Element Analysis*, 1st edn. McGraw-Hill: New York, 2003.
12. Boresi AP, Schmidt RJ, Sidebottom OM. *Advanced Mechanics of Materials*, 5th edn. John Wiley & Sons: New York, 1993.
13. Timoshenko SP, Woinovsky-Krieger S. *Theory of Plates and Shells*, 2nd edn. McGraw-Hill: New York, 1959.
14. Levy R, Spillers WR. *Analysis of Geometrically Nonlinear Structures*, Kluwer Academic Publishers: Dordrecht, The Netherlands, 2003; 2nd.
15. MSC Software. *MD Nastran R2.1 Quick Reference Guide*.
16. MSC Software. *MSC.Nastran 2004 Reference Manual*.
17. Weissstein EW. Archimedes' spiral. MathWorld—a Wolfram web resource. <http://mathworld.wolfram.com/ArchimedesSpiral.html>.
18. Wikipedia. Developable surface—wikipedia, the free encyclopedia. 2009. http://en.wikipedia.org/w/index.php?title=Developable_surface&oldid=230611228 [accessed 19 February 2009].
19. Wikipedia. Ruled surface—wikipedia, the free encyclopedia. 2009. http://en.wikipedia.org/w/index.php?title=Ruled_surface&oldid=270846259 [accessed 19 February 2009].
20. Kilian M, Flory S, Chen Z, Mitra NJ, Sheffer A, Pottman H. Curved folding. *ACM Transactions on Graphics* 2008; **27**(3): 1–9.
21. Oppenheim AV, Schaffer RW. *Digital Signal Processing*, Prentice-Hall Inc.: Englewood Cliffs, 1975.